

Mechanism and Modulation of Inactivation of the Kv3 Potassium Channel

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The mechanism and modulation of inactivation of the rat brain voltage-gated potassium channel, Kv3, expressed in *Xenopus* oocytes, were examined. In the cell-attached patch recording mode, Kv3 macroscopic current amplitude remains constant from one pulse to the next when a series of short (15 ms) repetitive depolarizing pulses are given. However, when the membrane patch is detached from the cell, the current amplitude decreases substantially from one depolarizing pulse to the next, even though there is no significant inactivation-like behaviour within a single current trace. In contrast, the closely related Kv2 potassium channel does not show a similar modulation of inactivation on changing from the cell-attached to the detached-patch recording mode. The data indicate the existence of a prolonged non-conducting state which the Kv3 channel enters in response to a series of short depolarizing pulses; this state is identical to the C-type inactivation state into which channels are driven during individual long depolarizing pulses. The probability of finding the channel in this state differs between the cell-attached and detached-patch modes, indicating that it is modulated by some cytoplasmic factor.

KEY WORDS: Channel modulation Inactivation Oocyte expression Patch-clamp Potassium channel

INTRODUCTION

Inactivation of voltage-dependent potassium (Kv) channels is divided into two types. Fast or N-type inactivation with a time course of a few milliseconds is well defined in terms of the mechanisms and structures involved in the process; it has been shown to be an open channel block by the N-terminal twenty amino acids of the channel protein (Hoshi *et al* 1990; Demo and Yellen 1991; Ruppersberg *et al* 1991). C-type inactivation (Hoshi *et al* 1991), which is the subject of the present report, is less well defined structurally and functionally. It is produced by a mechanism distinct from N-type inactivation (Choi *et al* 1991); however, no specific structure seems to account

entirely for C-type inactivation (Van Dongen *et al* 1990; Hoshi *et al* 1991; Busch *et al* 1991). Although traditionally C-type inactivation has been defined as a slow inactivation process, occurring over hundreds of milliseconds, that is observed during prolonged depolarizations, recent evidence suggests that, under some conditions, it can be very fast (Hoshi *et al* 1991).

Kv3 (Stühmer *et al* 1989; Swanson *et al* 1990) is a brain delayed rectifier Kv channel, highly selective for potassium, voltage-gated and slow to inactivate. When expressed in *Xenopus* oocyte membranes, it activates within a few milliseconds in response to depolarizing voltage steps, and inactivates over hundreds of milliseconds during prolonged depolarizations (Stühmer *et al* 1989; Swanson *et al* 1990). In this study we show that Kv3 in the detached inside-out patch recording mode can inactivate in response to a series of very short (15 ms) depolarizing voltage pulses, even though no

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inactivation is seen during each individual short depolarizing pulse. The results indicate that this type of short-pulse induced inactivation is equivalent in many of its properties to the slower C-type inactivation observed during more prolonged depolarizing pulses. Interestingly, the inactivation is modulated by some intracellular factor, since in the cell-attached recording mode the channel inactivates to a much lesser extent than in the detached-patch mode.

RESULTS

We expressed the voltage-dependent delayed rectifier channel Kv3 (Swanson *et al* 1990; Kv1.3 in the nomenclature of Chandy *et al* 1991) in *Xenopus* oocytes. Figure 1a shows that upon a very short (15 ms) depolarization in the cell-attached mode, virtually no inactivation is seen during a single current trace, and the current amplitude remains constant from one pulse to the next for 12 successive pulses (2 s inter-pulse interval). However, as demonstrated in Figure 1b, in all cases when the detached inside-out patch mode was used (>50 patches tested), a phenomenon is observed that at first glance resembles channel "rundown". Upon repetitive depolarizations of 15 ms duration, the Kv3 current amplitude in detached membrane patches decreased substantially from one pulse to the next, even though no inactivation-like be-

haviour was observed within a single current trace. We investigated the mechanism of this change in channel kinetics elicited by detaching the patch from the cell.

To investigate the specificity of this phenomenon we carried out the same experiment using the highly homologous Kv2 potassium channel (Swanson *et al* 1990; Kv1.6 according to Chandy *et al* 1991). As shown in Figure 2a, Kv2 resembles Kv3 in its lack of inactivation within a short depolarizing pulse, or from one pulse to the next, in the cell-attached mode. However, in contrast to Kv3, Kv2 current amplitude remains constant from one pulse to the next in the detached-patch mode (Figure 2b; compare Kv3, Figure 1b). The *Shaker-H4* mutant $\Delta 6-46$ (N-type inactivation removed) was also tested: no drop in current amplitude from one short depolarizing pulse to the next was observed in the cell-attached or detached-patch modes (data not shown). Thus, of the three Kv channels tested, only Kv3 exhibited modulation of inactivation when the membrane patch was detached from the cell. This specificity strongly suggests that the change observed on switching from the cell-attached to the detached-patch mode is not due simply to channel "rundown" or deterioration of the patch.

The rate of decrease of Kv3 current amplitude from one pulse to the next is related to the frequency of the repetitive voltage pulse. The relationship between the inter-pulse interval and

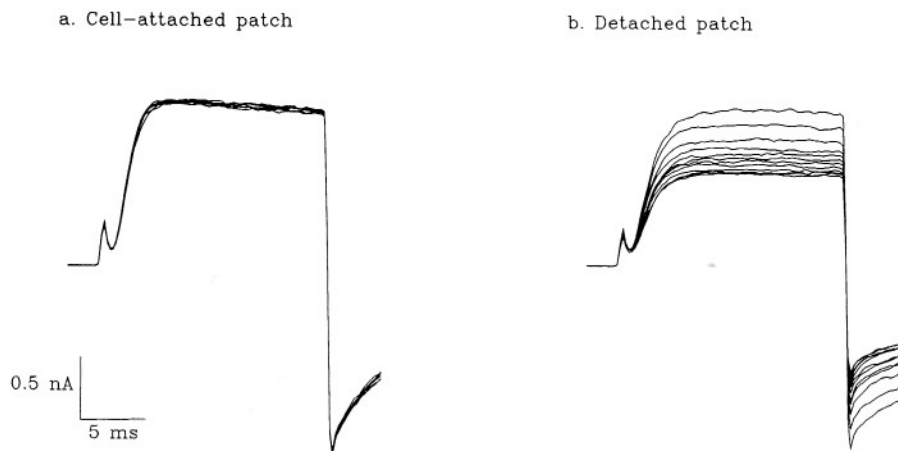


Figure 1 Decrease in the availability of Kv3 channels for opening upon changing from the cell-attached to the detached-patch mode. Patch recordings from Kv3-injected oocytes. (a) and (b) depict macroscopic patch currents evoked by 12 successive depolarizing voltage pulses, 2 s apart, from a holding potential of -90 mV to a pulse potential of $+40$ mV. Note that the currents are stable from one pulse to another on-cell (a), but decrease in the same patch off-cell (b). In both on-cell and off-cell modes, no inactivation is observed within each short pulse.

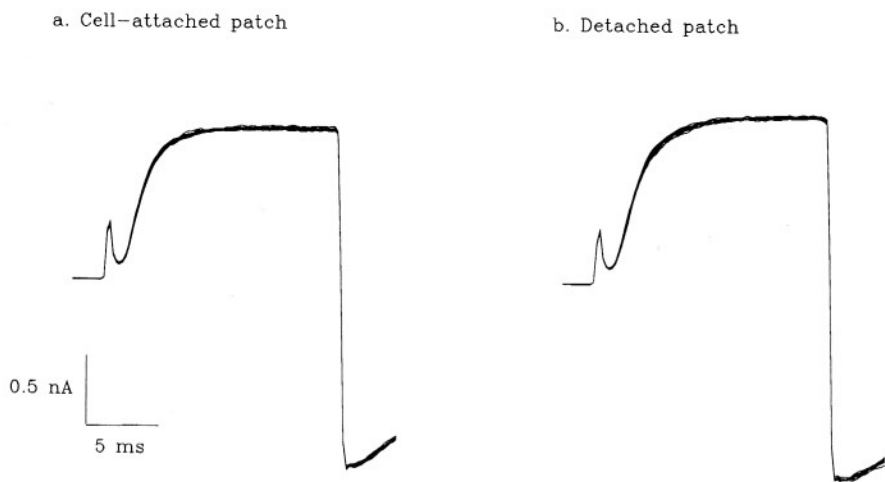


Figure 2 Stability of Kv2 currents upon changing from the cell-attached to the detached-patch mode. Similar to Figure 1, except that recordings were made from Kv2-injected oocytes. Currents evoked by 12 successive depolarizing pulses, 2 s apart, delivered to (a) cell-attached patches or (b) detached patches.

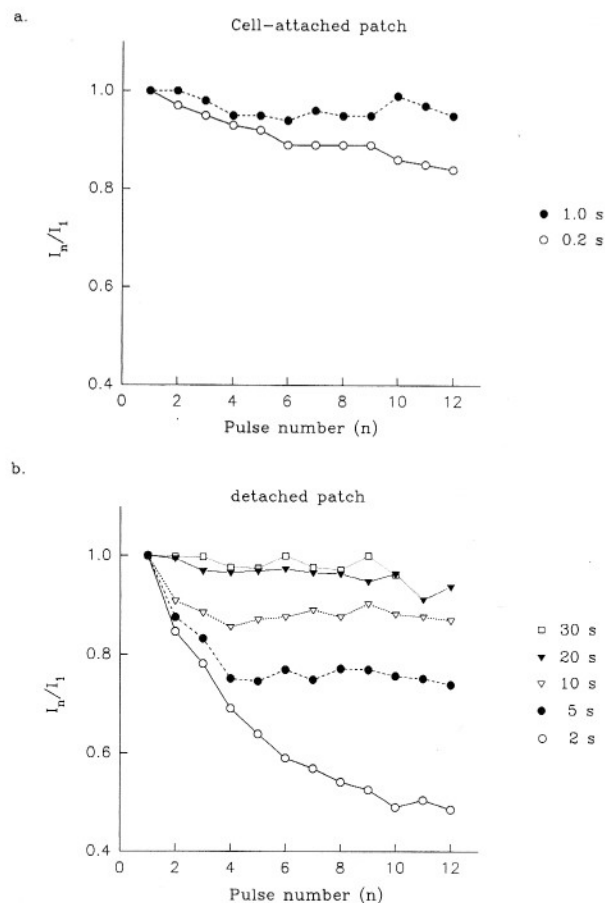


Figure 3 Kv3 current amplitude decrease from one repetitive voltage pulse to the next is related to the inter-pulse interval. Decrease of the normalized peak Kv3 current amplitude from one pulse to the next, as a function of inter-pulse interval, in the cell-attached (a) and detached-patch (b) recording modes. The numbers next to the traces are the inter-pulse intervals.

the current amplitude, in both the detached-patch and cell-attached modes, is shown in Figure 3. In the detached-patch mode, current amplitude decreases from one pulse to the next with inter-pulse intervals as long as 10 s (Figure 3b). In contrast, in the cell-attached patch mode, the current amplitude decreases from one depolarizing pulse to the next only if much shorter inter-pulse intervals are used (<1 s; Figure 3a). For any given inter-pulse interval, the decrement is far more pronounced in the detached-patch compared to the cell-attached mode. Note also that, even in the detached-patch mode, current amplitudes are stable from one pulse to the next if the inter-pulse interval is sufficiently long, further confirming that the change on switching from the cell-attached to the detached-patch configuration cannot be attributed to channel rundown.

The non-inactivating time course of the currents within a single trace, together with the "accumulation" of channels in the non-conducting state from one pulse to another, suggest that the channel can enter a non-conducting conformation, which we call the S state, either between the voltage pulses, or rapidly from a closed state (prior to channel opening) during the pulses (see Discussion and Aldrich 1981; DeCoursey 1990). The sets of experimental evidence, summarized below, indicate that the S state is identical to the C-type inactivation state (I_s) which channels enter during a long depolarizing pulse. These experiments also demonstrate that the differences observed between the cell-attached and detached-patch recording

modes result from modulation of the transition rates that control C-type inactivation of the Kv3 channel.

Recovery from S and Is follow the same time course Identical rates of recovery from S and Is are an important indicator of the identity of these states. The recovery rate constant from the S state can be estimated as shown in Figure 4 (filled circles), where the ratio of peak current amplitudes during the second and first of two short pulses, measured in a detached patch, is plotted at different inter-pulse intervals. The recovery of Kv3 from the S state can be described by a single exponential term with a time constant of 10 s. A similar double pulse protocol, but with a longer pulse duration (225 ms), was applied to test the rate of recovery from Is. This is not an ideal experiment because the recovery after a long pulse is likely to involve a mixture of recoveries from S and Is. In order to maximize the contribution of Is in these measurements, the ratio of current amplitudes at the ends of the pulse was taken as a measure of recovery from Is. As shown in Figure 4 (open circles), the recovery from Is can be described by a single exponential term having a time constant

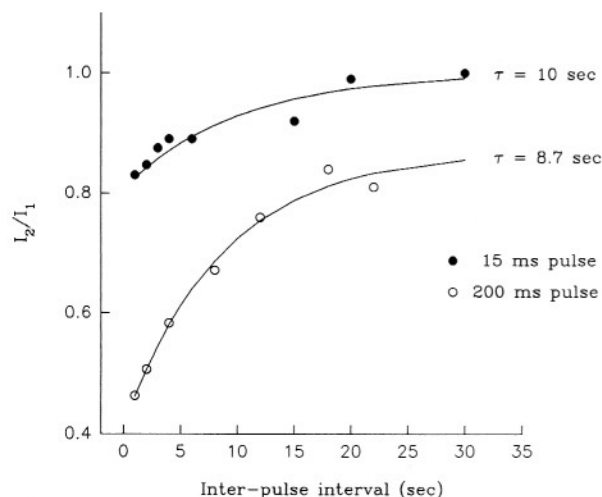


Figure 4 Recovery from the S state within a pulse follows the same time course as recovery from C-type inactivation. Filled circles are a plot of the ratio of peak Kv3 current amplitude during the second (I_2) and first (I_1) pulses of a pair of very short (15 ms) depolarizing pulses, measured in a detached patch at different inter-pulse intervals (see Figure 2) to allow estimation of the time course of recovery from the S state. The time constant for recovery is 10 s. Open circles are a similar two-pulse experiment, but with much longer (225 ms) pulses and measurement of the currents at the ends of the pulses to estimate the time course of recovery from C-type inactivation. Holding voltage -90 mV; pulse voltage $+40$ mV.

(8.7 s) that is very close to the time constant of recovery from the S state.

External tetraethylammonium (TEA) decreases the probability of finding Kv3 channels in the S and Is states External TEA has been used to interfere with slow inactivation of delayed-rectifier K^+ channels in lymphocytes (Grissmer and Cahalan 1989) and with C-type inactivation of *Shaker-H4* $\Delta 6-46$ (Choi *et al* 1991) and the cloned RKG5 potassium channel (Busch *et al* 1991) equivalent to Kv3 (Chandy *et al* 1991). These studies demonstrated that C-type inactivation of these channels is retarded in the presence of external TEA. As shown in Figure 5a, 30 mM external TEA also

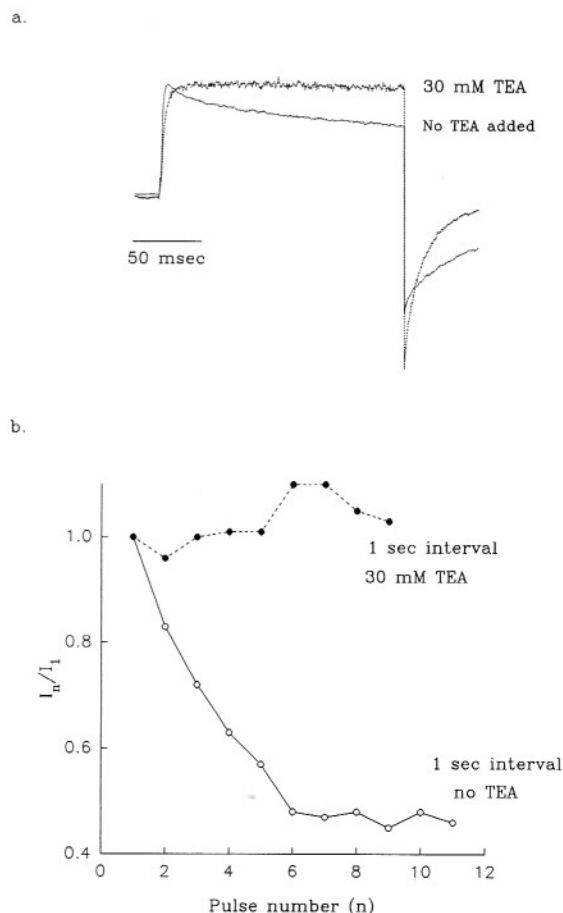


Figure 5 Extracellular TEA interferes with C-type inactivation of Kv3 channels, and with the probability of finding them in the S state. (a) Inactivation time course within a 200 ms pulse to $+40$ mV in the detached-patch mode, in the absence and presence of 30 mM TEA in the pipet (extracellular). Currents are scaled to the same peak level. (b) The decrease in current amplitude from one pulse to the next (probability of finding channels in the S state) is much smaller with 30 mM TEA in the pipet.

retards C-type inactivation of Kv3 during prolonged depolarizations in detached patches. Figure 5b details the effect of 30 mM TEA on the accumulation of channels in the S state in detached patches as a function of pulse frequency. Accumulation in the S state, measured as the decrease in current amplitude from one pulse to the next at an inter-pulse interval of 1 s, is similarly inhibited by external TEA.

Changing from the cell-attached to the detached-patch mode modulates both S and Is The probability of the Kv3 channel accumulating in the S state is increased by changing from the cell-attached to the detached-patch recording mode (Figure 1). If $S = I_s$, the accumulation in I_s during a prolonged depolarizing pulse should also be enhanced by removing the channel from its intracellular environment. As shown in Figure 6a, inactivation of Kv3 within a 200 ms pulse to +40 mV is indeed faster in the detached-patch than in the cell-attached mode.

We then investigated a possible correlation between the rate of C-type inactivation during long depolarizing pulses, and the decrease in current from one short depolarizing pulse to the next in other Kv channels. In Figure 6b the rates of C-type inactivation of Kv3 and of Kv2, a highly homologous channel (Swanson *et al* 1990), during prolonged depolarizing pulses are compared in the detached patch mode. C-type inactivation of the Kv2 channel is much slower than that of Kv3 at all voltages examined (see also Swanson *et al* 1990 and Kirsch *et al* 1991). This is consistent with the finding that Kv2 current amplitude does not fall significantly from one short depolarizing pulse to the next in the detached-patch mode (Figure 2b; compare with Kv3, Figure 1b). The *Shaker-H4* mutant $\Delta 6-46$ also demonstrates much slower C-type inactivation (Hoshi *et al* 1991) than the Kv3 channel and does not exhibit a drop in current from one rapid short pulse to the next (data not shown). Thus, for three different Kv channels, the decrease of current from one short pulse to the next is correlated with the rate of C-type inactivation during prolonged depolarization.

An initial search was carried out for the channel structural features that underlie the modulation of inactivation kinetics in the Kv3 channel. In spite of their high homology, Kv2 and Kv3 have very different cytoplasmic C-terminus domains, including different consensus sites for phosphorylation

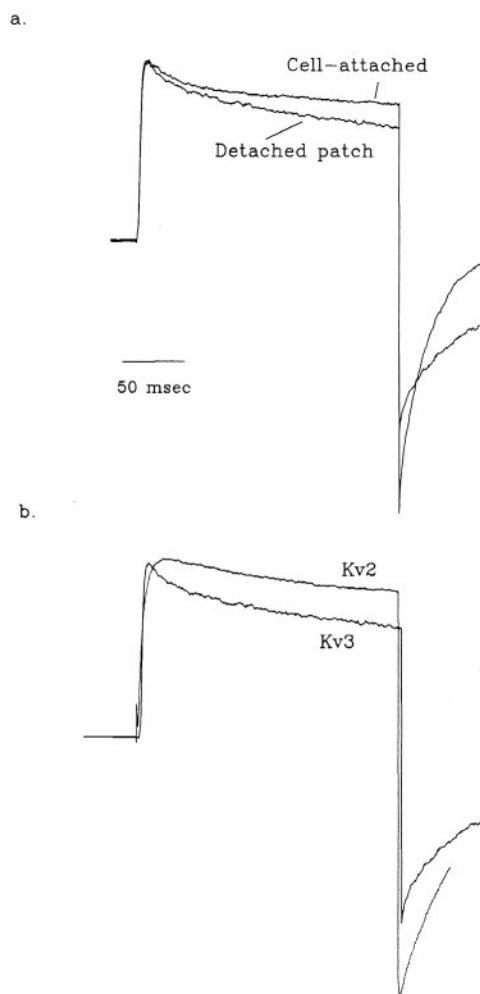


Figure 6 C-type inactivation of Kv3 and Kv2. (a) Inactivation time course of Kv3 current within a prolonged depolarizing pulse in the cell-attached and detached-patch modes, recorded from the same patch. The peak currents in the two recording modes are normalized to the same value. C-type inactivation of Kv3 is faster in the detached than in the cell-attached patch. (b) Comparison of C-type inactivation time courses of the Kv3 and Kv2 channels. Superposition of scaled current traces evoked in the detached-patch mode from a Kv3 injected oocyte, and a Kv2 injected oocyte. C-type inactivation of Kv3 channels during a prolonged pulse is faster than that of Kv2. For both (a) and (b): holding potential -90 mV, command potential $+40$ mV, pulse duration 200 ms.

(Swanson *et al* 1990). These C-terminus sequence differences led us to test a chimeric Kv2 channel construct bearing a Kv3 cytoplasmic C-terminus. This channel behaves like the parent Kv2 channel with regard to C-type inactivation in the cell-attached mode (as expected from Busch *et al* 1991), without modulation of inactivation kinetics on changing to the detached-patch mode (Table 1). The "reversed" chimeric channel (Kv2 cytoplasmic

Table 1 Results from chimeric channels indicate that the C-terminus is not involved in modulation of inactivation kinetics upon detaching the patch from the cell. Kv2 and Kv3 channels and chimeric channel constructs containing portions of Kv2, Kv3 and *Shaker-H4* $\Delta 6-46$ were expressed in oocytes, and patch recording was carried out. Twelve voltage pulses to +40 mV, 1 s apart, were applied to the patch in the cell-attached and detached-patch recording modes. The ratio of peak current amplitude of the last pulse over the first pulse is given ($\pm$ SD). *n*, number of patches tested

Channel construct	Cell-attached patch	Detached patch
Kv3	0.89 $\pm$ 0.04 (<i>n</i> = 4)	0.56 $\pm$ 0.10 (<i>n</i> = 4)
Kv2	0.99 $\pm$ 0.03 (<i>n</i> = 3)	0.97 $\pm$ 0.05 (<i>n</i> = 3)
Kv3 backbone + Kv2 C-terminus	0.97 $\pm$ 0.01 (<i>n</i> = 3)	0.46 $\pm$ 0.13 (<i>n</i> = 3)
Kv2 backbone + Kv3 C-terminus	1.03 $\pm$ 0.03 (<i>n</i> = 2)	1.00 $\pm$ 0.05 (<i>n</i> = 3)
<i>Shaker</i> backbone + Kv3 C-terminus	1.00 $\pm$ 0.03 (<i>n</i> = 3)	1.03 $\pm$ 0.05 (<i>n</i> = 3)

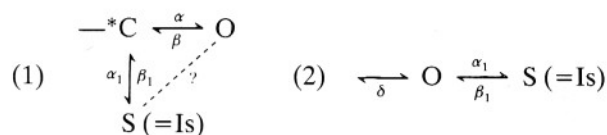
C-terminus spliced onto Kv3) behaves like the parent Kv3 channel with regard both to C-type inactivation in the cell-attached mode and to modulation of inactivation kinetics on changing to the detached-patch mode (Table 1). Furthermore, when the Kv3 C-terminus is spliced onto the *Shaker-H4* $\Delta 6-46$ channel, modulation is again absent after the patch is detached from the cell (see Table 1). These experiments indicate that the cytoplasmic C-terminus cannot, by itself, account for the modulation observed on changing from the cell attached to the detached-patch recording modes.

DISCUSSION

Following repetitive short (<15 ms) membrane depolarizations the Kv3 potassium channel enters a prolonged non-conducting state (S), even though no inactivation is observed within each pulse. The probability of finding the channel in this state is related to a still unidentified cytoplasmic factor. Both the S state, and C-type inactivation (Is), which develops during long depolarizing pulses, behave similarly with respect to recovery rates, response to external TEA and modulation by the cytoplasmic factor. Accordingly, we conclude that the S state is identical to the Is state.

Kinetic models for the slow inactivation of Kv channels range from those that require the complete coupling of inactivation to prior channel opening (e.g. Grissmer and Cahalan 1989; Iverson and Rudy 1990; Koren *et al* 1990) to more relaxed

models where slow inactivation can branch from any one of several states along the activation pathway (e.g. Hoshi *et al* 1991). In the case of Kv3, the lack of inactivation within the depolarizing voltage pulse, together with the "accumulation" of channels in the non-conducting state from one pulse to another, can be explained by the existence of a non-conducting state that branches from a closed state before the channel has a chance to open. Kinetic scheme 1 outlines such a model, with rate constants α , α_1 , β and β_1 such that $\beta \gg \beta_1$ and $\alpha_1 \gg \beta_1$; *C represents a set of closed states from which the S state can branch; Is stands for a C-type inactivation state that is coupled to the open (O) state. This scheme is similar to one suggested by Aldrich (1981) to account for cumulative inactivation of potassium currents in molluscan neurons: that is, for sufficiently short pulses, C-type inactivation within a pulse (classically thought of as an O to Is transition) is avoided completely, as evidenced by the non-decrementing time course of the current within the pulse (Figure 1).



Kinetic scheme 2 predicts equally well the observed decrement in current amplitude between depolarizing voltage steps, provided that the voltage-dependent deactivation rate (δ) is either slower, or not much larger, than the voltage-independent inactivation rate constant (α_1). Given these constraints, the channel can inactivate from an open state during the long deactivation phase between the voltage pulses. Kinetic scheme 2 was shown by DeCoursey (1990) to predict cumulative inactivation in type II alveolar epithelial cell potassium channels. A careful examination of Figure 1 and Figure 6a shows that the tail currents are significantly slower in the detached-patch as compared to the cell-attached mode, which is consistent with such a scheme.

The existence of a prolonged non-conducting state is not unique to Kv3. Koren *et al* (1990) identified a prolonged closed state following the unitary inactivation transition of a single mammalian Kv channel. On the basis of different recovery rates, they concluded that this absorbing state is not identical to the inactivated states entered by channels during long depolarizations.

Potassium channels found in PC-12 cells (Hoshi and Aldrich 1988) also show a tendency to enter an absorbing state kinetically distinct from the interburst closures that represent slow inactivation. In both cases it was hypothesized that the absorbing inactivated state is coupled sequentially to the open state (as suggested in kinetic scheme 2).

Which of the two kinetic schemes, outlined above, more accurately describes the inactivation of Kv3 channels remains to be determined. Single channel recordings should resolve this question. Both schemes are intriguing since they direct attention to processes that take place between depolarizing voltage pulses (or, under physiological conditions, between action potentials) as determinants of the rate of inactivation. Furthermore, both schemes imply that the so-called "slow" C-type inactivation of the Kv3 channel can in fact occur very rapidly (see also Hoshi *et al* 1991). Whether the ability to enter a prolonged inactivated state after a very short depolarizing pulse is a common feature of all Kv channels is not yet clear. The Kv3 channel differs from most other members of its family in at least two respects: it exhibits one of the fastest C-type inactivations (Stühmer *et al* 1989), and its recovery from C-type inactivation is very slow. On the other hand, these properties in Kv3 might simply represent an extreme form of a mechanism that exists in many other delayed rectifier potassium channels.

In contrast to the relatively slow C-type inactivation that develops during a pulse, an inactivation process like that described here would have a marked effect on firing rate and the duration of action potentials. Modulation of this process may therefore be of physiological importance. The change in the probability of finding the Kv3 channel in the inactivated state produced by switching from the cell-attached to the detached-patch mode is intriguing. We believe that this may reflect a physiologically relevant modulatory mechanism for the following reasons. First, the phenomenon described here is not restricted to oocytes. Cahalan *et al* (1985) described a similar phenomenon in a voltage-gated potassium current in human T lymphocytes while recording in the whole-cell mode. Interestingly, recent data suggest that the Kv3 channel is expressed in T lymphocytes (Grissmer *et al* 1990). Second, as shown here, the phenomenon is not channel "rundown", but rather results from a change in one or more kinetic rate constants on switching to the off-cell mode. This

fact argues strongly in favour of the involvement of a cellular component in modulating these kinetic parameters.

The modulation described here is specific to Kv3 and is not exhibited by the highly homologous Kv2 channel. These two cloned potassium channels are closely related structurally, yet reveal very different patterns of modulation, and thus provide a tool for elucidating the detailed molecular mechanisms of these channel modulatory phenomena. The different C-type inactivation kinetics seen in the Kv3 and Kv2 channels during a prolonged depolarizing pulse can be accounted for by a single histidine-to-tyrosine change in position 401 of Kv3 (Busch *et al* 1991). We are constructing this mutant to determine whether inactivation kinetics is modulated. Other studies (Hoshi *et al* 1991) have implicated the S6 region in C-type inactivation, but this cannot account for the modulation observed here as Kv2 and Kv3 have identical S6 regions (Swanson *et al* 1990). Furthermore, the chimeric channel experiments described here demonstrate that the cytoplasmic C-terminus cannot by itself account for the modulation. These experiments, together with the identity of Kv2 and Kv3 channels in their S6 regions, indicate that structures upstream from the C-terminus and S6 participate in the modulation of Kv3 inactivation.

MATERIALS AND METHODS

Expression of Kv channels in oocytes

Kv3 and Kv2 genes in pGEM-A, a modified version of Promega pGEM-9zf(-), were kindly provided by Dr. Richard Swanson (Swanson *et al* 1990). The constructs were modified by additions of an Nco1 site at the initiation ATG, and a BamH1 site at the beginning of the S6 region, without affecting the amino acid sequence. To enhance expression of the channels in *Xenopus* oocytes, their native 5' untranslated regions (5' UT) were replaced with the 184 base pair sequence preceding the coding domain of *Shaker-H4* (Steve A. N. Goldstein, unpublished results). *Shaker-H4* Δ 6-46 was kindly provided by Dr. Roderick MacKinnon in pBluescript (Yellen *et al* 1991), and was manipulated to facilitate the isolation of its 5' UT domain and construction of *Shaker/Kv* chimeras by addition of an Nco1 site at the initiation ATG. Standard methods were used for DNA and RNA

preparation (Sambrook *et al* 1989). T7 RNA polymerase was used for transcription after plasmid linearization with NotI (Kv2 and Kv3) or HindIII (*Shaker-H4* Δ 6-46). Oocytes were harvested from mature *Xenopus laevis* females and dissociated in 2 mg/ml collagenase (1A, Sigma). Stage V–VI defolliculated cells were injected with $\sim$ 40 nl mRNA ($\sim$ 1 mg/ml). Injected cells were incubated at 18°C for up to 7 d in frog Ringer's solution. Expression levels are such that within 72 h of injection, we routinely see 0.5–10 nA of Kv3 and 0.1–2 nA of Kv2 current per patch.

To construct chimeric Kv3/2 and *Shaker*/Kv3 channels, the parent plasmids were digested with NcoI and BamHI, and both backbones and inserts were cut out of a gel and purified using a GeneClean kit (Bio 101) or a paper purification procedure (Sambrook *et al* 1989). The resulting pieces were ligated (T4 ligase) to create a Kv3 channel with a Kv2 C-terminus, a Kv2 channel with a Kv3 C-terminus, and a *Shaker-H4* channel with a Kv3 C-terminus. Each construct was sequenced across the BamHI site using a Circum Vent (New-England Biolab) PCR sequencing kit.

Electrophysiological recordings and data analysis

Macroscopic currents were recorded in cell-attached and detached inside-out membrane patches from devitellinized oocytes, using an Axopatch 200 or List EPC/7 patch clamp amplifier. The amplifier output was filtered at 1–3 kHz (8-pole Bessel filter, Frequency Devices, Haverhill, MA), digitized at 10 kHz and stored for later analysis. Data traces used for analysis were digitally subtracted for leak and capacitance. Single exponential fits were computed using the steepest descent method to minimize sums of squares. Unless otherwise mentioned in the text, the bath solution contained (mM): 100 KCl, 1 EGTA and 10 Hepes, pH 7.5; pipet solution contained (mM): 30 KCl, 70 NaCl, 2 CaCl₂ and 10 Hepes, pH 7.5. Under these conditions, and by analysing data from patches with <2.5 nA of peak current, we were

able to avoid the adverse affects of potassium accumulation in an unstirred layer at the tip of the recording pipet. All experiments were carried out at room temperature.

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